Function 11: TIFF/JPG Representations — Full API Method

Last Updated: March 25, 2026
Branch: `Function-11-API`
Function Status: Active — Full API approach (no Digital Uploader required)
Based on: Harvard/Corinna LibraryTechServices spec + Alma Representations REST API
Archive of previous approach: `FUNCTION_11_PREPARE_TIFF_JPG_CSV-Method-Archive.md`

Overview

Function 11 creates DERIVATIVE_COPY JPEG representations on Alma bib records and populates them with JPG derivatives converted from local TIFF files — **entirely via the Alma API and ExLibris S3 bucket**. No manual Digital Uploader steps, no local output directory, no `values.csv`.

What it does, end-to-end

For each MMS ID in the loaded set:

1. Look up the TIFF filename from the CSV file specified in `TIFF_CSV_FILE` environment variable
2. Search for the TIFF file recursively under `LOCAL_TIFF_PATH` (or download from S3 if not found locally)
3. Convert the TIFF to a JPG (in a temporary directory, auto-cleaned)
4. Upload the JPG to the ExLibris-managed AWS S3 bucket under `{institution_code}/upload/`
5. Check the Alma bib record for an existing DERIVATIVE_COPY / JPG representation:
    - **Rep has a file already** → **SKIP** the record. Log the state, make no changes.
    - **Rep exists but is empty** → POST the new JPG file into it
    - **No matching rep exists** → POST to create a new DERIVATIVE_COPY rep, then POST the JPG file
6. Temporary JPG files are automatically deleted when the batch finishes

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Prerequisites

Required `.env` variables

Add these to the `.env` file in the CABB workspace (see `.env.example` for the full template):

```
ALMA_API_KEY=your_alma_api_key
ALMA_API_REGION=America
ALMA_INSTITUTION_CODE=01GCL_INST            # Your institution code
ALMA_S3_BUCKET=na-st01.ext.exlibrisgroup.com        # Production
# ALMA_S3_BUCKET=na-test-st01.ext.exlibrisgroup.com  # Sandbox

AWS_ACCESS_KEY_ID=your_exl_s3_access_key
AWS_SECRET_ACCESS_KEY=your_exl_s3_secret_key
AWS_DEFAULT_REGION=us-east-1

TIFF_CSV_FILE=single_tiff_objects_20260320_121506.csv

LOCAL_TIFF_PATH=/Volumes/Acasis1TB
```

> **Getting S3 credentials:** The ExLibris S3 credentials are separate from any personal AWS credentials. Request them from your Alma/ExLibris administrator. They are scoped to the institution's upload bucket only.

Required Python libraries

```
source .venv/bin/activate
pip install Pillow boto3
```

Both are verified at runtime; the function will stop immediately with a clear error message if either is missing.

Required CSV file

The CSV file specified in `TIFF_CSV_FILE` environment variable must exist in the CABB workspace root.

Column	Description	Required
`MMS ID`	Alma bibliographic record ID	Yes
`TIFF Filename`	Filename only (e.g., `photo_001.tif`)	Yes
`S3 Path`	Full S3 bucket key path for fallback download	Optional

```
MMS ID,TIFF Filename,S3 Path
991234567890104641,photo_001.tif,01GCL_INST/sync/collection1/photo_001.tif
991234567890204641,photo_002.tiff,01GCL_INST/sync/collection2/photo_002.tiff
```

How file location works:

- The function searches for `TIFF Filename` recursively under the path specified in `LOCAL_TIFF_PATH` environment variable
- If found locally, it uses the local file
- If NOT found locally AND `S3 Path` is provided, it downloads from the ExLibris S3 bucket as a fallback
- If neither local file nor S3 path is available, the record fails

Notes:

- Lines starting with `#` are treated as comments and skipped
- Both `.tif` and `.tiff` extensions are supported
- `S3 Path` format: `{institution_code}/sync/{collection}/{filename}` or similar bucket structure

Processing Logic

Decision tree per record

```
MMS ID received
  |
```

```
  |-- Not in TIFF CSV                    --> record as "no_path"
(no_path_count++)
  |-- TIFF file not found locally        --> Try S3 fallback download
  |     |                                       |
  |     |-- S3 path missing/download fails --> record as "no_file"
(no_file_count++)
  |     |-- S3 download succeeds           --> Continue
  |
  |-- Convert TIFF --> JPG               <-- fails? --> FAIL (failed_count++)
  |-- Upload JPG to ExL S3               <-- fails? --> FAIL
  |
  |-- GET existing representations from Alma API
        |
        |-- DERIVATIVE_COPY/JPG rep found WITH file
        |       --> SKIP (skipped_has_file_count++)
        |         log: "SKIPPED -- existing rep already has file (pid: ...)"
        |
        |-- DERIVATIVE_COPY/JPG rep found, EMPTY
        |       --> POST file to existing rep --> SUCCESS (success_count++)
        |
        +-- No matching rep found
                --> POST create new DERIVATIVE_COPY rep
                --> POST file to new rep --> SUCCESS (success_count++)
```

What "matching representation" means

The code searches the GET response for a representation where:

- `usage_type.value == "DERIVATIVE_COPY"`, AND
- `label` contains `"JPG"`, `"jpg"`, or `"derivative"` (case-insensitive)

If the inline file list in the GET response is ambiguous, a second `GET /representations/{rep_id}/files` call is made to confirm whether a file exists before deciding to skip.

How to Use

Step 1: Configure `.env`

Ensure all required variables are set. The function stops immediately with a clear error message if `ALMA_S3_BUCKET` or the AWS credentials are missing.

Step 2: Verify prerequisites

```
# Check CSV exists (use filename from your TIFF_CSV_FILE env variable)
ls -lh single_tiff_objects_20260320_121506.csv

# Check libraries
source .venv/bin/activate
python -c "from PIL import Image; import boto3; print('OK')"
```

```
# Check volumes are mounted (if using network paths)
ls /Volumes/DGIngest/
```

Step 3: Load a set in CABB

- Enter a **Set ID** and click **"Load Set Members"**, OR
- Enter a single **MMS ID** in the MMS ID field for single-record mode

Step 4: Run Function 11

1. Select **"11 - Prepare TIFF/JPG Representations"** from the function dropdown
2. Click **"Run Function on Set"**
3. Review the confirmation dialog — it confirms the API+S3 approach and credential requirements
4. Click **"Proceed"**
5. Monitor the log window (progress bar updates after each record)

Step 5: Review log output

The log shows a detail line for every record processed. At the end, the summary line reads:

```
Function 11 (API method) complete:
  N succeeded,
  N skipped (rep already has file),
  N failed,
  N missing path,
  N file not found,
  N retrieved from S3 fallback
```

If `success_count > 0`, the files are already live in Alma — **no further upload steps needed**.

Step 6: Verify in Alma

- Open a processed record in Alma
- Navigate to **Resources > Manage Inventory > Manage Digital Files**
- Confirm the DERIVATIVE_COPY representation has a JPG file attached
- (Optional) Check the Primo/Discovery view after ~15 minutes for the thumbnail

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Output

There is **no local output directory**. All JPG conversion files are written to a `tempfile.TemporaryDirectory` that is automatically cleaned up when the batch finishes. Nothing is written to `~/Downloads/` or any persistent path.

The only permanent output is the file attached to the Alma digital representation.

Summary of API Calls Per Record

#	Method	Endpoint	Purpose
1	`GET`	`/almaws/v1/bibs/{mms_id}/representations`	Check for existing DERIVATIVE_COPY rep
2	`GET`	`/almaws/v1/bibs/{mms_id}/representations/{rep_id}/files`	Confirm file presence (conditionally)
3	`POST`	`/almaws/v1/bibs/{mms_id}/representations`	Create new rep (only if none found)
4	`POST`	`/almaws/v1/bibs/{mms_id}/representations/{rep_id}/files`	Attach JPG via S3 path reference

The S3 upload precedes call #4: a boto3 `put_object` writes the JPG to:

```
s3://{ALMA_S3_BUCKET}/{ALMA_INSTITUTION_CODE}/upload/{mms_id}.jpg
```

The file POST body that references the S3 path:

```
{
  "label": "991234567890104641.jpg",
  "path": "01GCL_INST/upload/991234567890104641.jpg"
}
```

Alma moves the file from the `/upload` staging area to `/storage` automatically.

Image Conversion Details

JPG output specifications:

- Format: JPEG, quality=95, optimize=True
- Color mode: RGB (all source modes normalized)
- Bit depth: 8-bit output

Mode handling in `_convert_tiff_to_jpg()`:

Source mode	Action
`I`, `I;16`, `I;16B`, `I;16L`, `I;16N`	Scale to 8-bit (pixel / 256), convert L then RGB
`RGBA`, `LA`	Composite alpha over white background, convert to RGB
`P` (palette)	Convert to RGBA first, then composite over white
`L` (grayscale)	Convert to RGB directly

Source mode	Action
`RGB`	Used as-is
Other	Force convert to RGB

Temporary JPG files are named `{mms_id}.jpg` during processing and deleted automatically.

Error Reference

Message in log	Cause	Fix
`API Key not configured`	`ALMA_API_KEY` missing from `.env`	Add it
`TIFF_CSV_FILE not configured`	Neither `tiff_csv` parameter nor `TIFF_CSV_FILE` env variable is set	Add `TIFF_CSV_FILE` to `.env`
`LOCAL_TIFF_PATH not configured`	Missing `.env` var	Add `LOCAL_TIFF_PATH=/Volumes/...` to `.env`
`ALMA_S3_BUCKET not configured`	Missing `.env` var	Add `ALMA_S3_BUCKET=...`
`AWS credentials not configured`	Missing `AWS_ACCESS_KEY_ID` or `AWS_SECRET_ACCESS_KEY`	Add both to `.env`
`TIFF CSV not found`	CSV file specified in `TIFF_CSV_FILE` not found	Check `TIFF_CSV_FILE` in `.env`
`Pillow library not installed`	Missing dependency	`pip install Pillow`
`boto3 library not installed`	Missing dependency	`pip install boto3`
`No record found in {csv}`	MMS ID not in TIFF CSV	Add row to CSV
`not found under {LOCAL_TIFF_PATH}`	TIFF filename not found locally	Mount volume or ensure S3 Path is correct
`No S3 path available for fallback`	TIFF not found locally and no S3 Path in CSV	Add S3 Path column or fix local path
`TIFF/JPG conversion failed`	Corrupt or unreadable TIFF	Verify TIFF integrity
`S3 upload failed: ...`	Wrong AWS credentials / bucket unreachable	Check `.env` AWS vars
`Failed to create representation: HTTP`	API auth failure or bad MMS ID	Check API key and MMS ID

Message in log	Cause	Fix
`4xx`		
`Failed to post file to representation`	S3 file not visible or wrong path	Check S3 key format
`SKIPPED -- existing rep already has file`	Rep already populated	Intentional — no changes made

FAQ

Q: Can I re-run Function 11 on records that were already processed?
Yes, but those records will be **skipped** automatically. The function detects the representation already has a file and leaves it untouched. Only records with empty or missing representations are processed.

Q: Can I process a single record without loading a set?
Yes. Enter the MMS ID in the MMS ID field instead of loading a set.

Q: What S3 bucket should I use for testing?
Use the sandbox bucket: `na-test-st01.ext.exlibrisgroup.com`. Set `ALMA_S3_BUCKET` to this in `.env` when testing against your Alma sandbox. Ensure you're also using the sandbox API key.

Q: Do I need the old Alma Digital Import Profile (ID 7848184990004641)?
No. That profile was for the previous CSV/Digital-Uploader approach. It is not used by Function 11.

Q: What if I need to change the JPG quality?
Edit `_convert_tiff_to_jpg()` in `app.py` and change `quality=95`. Valid JPEG quality range is 1-95.

Q: What if my TIFFs are spread across multiple directories under LOCAL_TIFF_PATH?
That is fine. The function searches recursively for the filename. If `LOCAL_TIFF_PATH` is `/Volumes/Acasis1TB`, it will find `photo.tif` whether it's at `/Volumes/Acasis1TB/collection1/photo.tif` or `/Volumes/Acasis1TB/deep/nested/folder/photo.tif`.

Q: What is the S3 Path used for?
The S3 Path is a fallback mechanism. If the TIFF filename isn't found under `LOCAL_TIFF_PATH` (e.g., volume not mounted, file moved), the function will attempt to download the TIFF from your institution's ExLibris S3 bucket using the S3 Path. This ensures processing can continue even if local files are temporarily unavailable.

Q: Will this touch bibliographic metadata?
No. Function 11 only reads, creates, and attaches files to digital representations. Bibliographic metadata is never modified.

Q: What if an S3 upload succeeds but the file POST fails?
The orphaned file in `/upload` will be cleaned up by ExLibris automatically. The record is marked as failed in the summary, and you can safely re-run it.

Related Documentation

File	Contents
`.env.example`	All required environment variables with comments
`FUNCTION_11_PREPARE_TIFF_JPG_CSV-Method-Archive.md`	Archived docs for the old CSV/Digital-Uploader approach
`FUNCTION_11_MIGRATION_NOTES.md`	History of prior CSV to XML migration (superseded)
`FUNCTION_11_CSV_PROFILE_SETUP.md`	Setup guide for old Alma Digital Import Profile (no longer needed)

Related Functions

- **Function 18** — Identify single-TIFF records (generates TIFF CSV file)
- **Function 14a** — Prepare Thumbnails (similar API-driven workflow, uses `AUXILIARY` usage type)